CS 516

PROJECT MILESTONE #2

"Representation Bias in Data: A Survey on Identification and Resolution Techniques"

-- by Nima Shahbazi, Yin Lin et al.

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OVERVIEW

- Introduction
- Causes
- Existing Literature
- Why does this matter?
- Measuring Bias
- Identifying RB in Structured Data
- Identifying RB in Unstructured Data
- Conclusion

INTRODUCTION

- Representation Bias: parts of the target population are under-represented in the dataset used to train or evaluate models, so the model will likely generalize poorly for those parts.
- data-centric problem, not an algorithmic one
- model can have high overall accuracy while still failing systematically on under-represented subgroups
- errors for minority groups are not just statistics; they can cause serious harm
- “found data” and lack of control: data scientists rarely control collection; they often work with “found data”; without a systematic data-collection plan that explicitly targets representativeness, representation bias is almost guaranteed
- many fairness techniques assume you can collect more/balanced data, but in practice you often can’t: this is what sets this paper apart

NOTE: Representation Bias has been studied before, this survey is unique because most fairness work focuses on model-level fairness constraints, whereas this survey focuses on bias inherent to datasets, irrespective of downstream models.

CAUSES

- Historical Bias: society's past decisions shaped what was recorded; cognitive biases in data collectors influence what gets captured
- Selection Bias/Sampling Bias: the way sampling is done excludes sub-populations; selection artifacts
- Skewed underlying distribution: random sampling when some groups are truly small in the population
- Post-collection biases: during data cleaning, labeling, aggregation, or feature engineering steps

EXISTING LITERATURE

Why survey fairness before bias:

- Fairness defines what “unbiased” means before we start fixing data or models.
- It sets the ethical and mathematical goals for later mitigation.

Fairness Types:

- Individual Fairness: similar individuals → similar outcomes
- Group Fairness: demographic groups treated equally
- Subgroup Fairness: captures smaller or hidden overlapping groups

Group Fairness Metrics:

- Statistical Parity – same positive outcome rate across groups
- Equalized Odds – equal error rates
- Predictive Parity – equal accuracy across groups

FAIRNESS MITIGATION TECHNIQUES

PRE-PROCESSING (Data Level)

- Fix bias before training
- Re-sampling or re-weighting to balance groups
- Remove sensitive attributes (e.g., gender, race)
- Goal: give model balanced, fair input

IN-PROCESSING (Model Level)

- Add fairness during training
- Apply fairness constraints to loss function
- Use adversarial debiasing to hide sensitive info
- Goal: make fairness part of learning

POST-PROCESSING (Output Level)

- Fix bias after training
- Adjust thresholds / re-calibrate outputs
- Equalize false-positive & false-negative rates
- Goal: fair final predictions

REPRESENTATION BIAS IN MACHINE LEARNING

Understanding how unequal data representation leads to unfair AI systems and real-world harm

What Is Representation Bias and Why It Matters

The Core Problem:

Representation bias occurs when certain sub-populations are unequally or insufficiently represented in training data, leading to unfair model behavior. The principle is simple yet critical: "Bias in → Bias out" — even the most sophisticated algorithms fail when trained on skewed data.

Real-World Consequences:

- Facial recognition systems misidentify individuals with darker skin tones.
- Medical diagnosis tools trained on datasets missing minority demographics.
- Hiring algorithms perpetuating existing workplace inequalities.

Historical Bias
Real-world inequality reflected in data (e.g., few female CEOs)

Distribution Skew
Natural population imbalance in collected samples

Selection Bias
Flawed or voluntary data collection methods

Critical insight: Historical bias cannot be fixed by better sampling alone — the "truth" being sampled is already unfair.

Measuring Representation Bias

Quantitative tools enable us to detect and measure bias systematically across datasets.

Representation Rate (τ)

Formula: $\tau = ni / nj$

Measures **global group** balance across the entire dataset. A low τ value indicates high representation bias. However, achieving $\tau = 1$ becomes challenging when the underlying population itself exhibits skew.

Data Coverage (k)

Definition: A Subgroup is "covered" if $\geq k$ similar samples exist nearby.

Detects **local gaps** in representation, particularly for intersectional groups. The k parameter is user-defined and depends on the dataset size and domain confidence requirements.

τ = Global Snapshot
Provides overall balance across major demographic groups

k = Local Zoom-In
Reveals granular coverage gaps in specific subgroups.

Visualization tip: Think of coverage as a heat map — red zones indicate regions with few data points, revealing bias hotspots.

WHERE BIAS HURTS MOST

Not all missing data creates equal harm. The location of bias matters more than its quantity. Bias near decision boundaries causes maximum damage, while gaps in irrelevant regions have minimal impact.

Decision Boundaries
Missing samples here lead to severe misclassification

Background Regions
Gaps in irrelevant areas cause minimal effect

The Cat-Ear Example:
In image classification: pixels missing near a cat's ear region → classifier mislabels ears. Pixels missing in background → no meaningful effect on classification accuracy.

Key Takeaway: Evaluate the spatial distribution and context of bias, not merely its aggregate quantity.

STRUCTURED DATA TAXONOMY

A comprehensive framework organizing bias detection and mitigation approaches across different data types and contexts.

Identification - Methods to find and measure bias.

Resolution - Techniques to fix and mitigate bias.

Attribute Type

- **Discrete:** Categorical variables (gender, race, occupation).
- **Continuous:** Numeric variables (age, income, scores)

Data Structure

- **Single table:** Unified dataset
- **Multi-relation:** Connected databases with foreign keys

Resolution Capability

- **With new data:** Collect additional samples.
- **No new data:** Re-weight or augment existing samples.

This taxonomy serves as your roadmap — every bias mitigation method discussed next fits somewhere within this structure.

IDENTIFYING REPRESENTATION BIAS IN STRUCTURED DATA

we examine 2 cases:

⇒ discrete attributes (categorical) → single relation, multiple relations

⇒ continuous attributes (numerical)

Identification Methods for Discrete Attributes

Multiple algorithmic approaches detect under-representation in categorical data, each offering unique perspectives on bias.

01

Pattern-Based Coverage

Find under-covered attribute combinations using algorithms like **Pattern-Breaker** and **Deep-Diver** to identify Maximal Uncovered Patterns (MUPs)

02

Slice-Based Analysis

SliceFinder: Ranks slices by effect size = $|\mu_s - \mu^D| / \sigma^D$ to locate poor model performance

SliceLine: Uses linear-algebra search for top-k problematic slices

03

Causal & Dependency Methods

Pradhan et al.: Remove patterns and retest fairness metrics

Azzalini et al.: Apply Conditional Functional Dependencies (CFDs)

04

Explainability Approaches

Divergence metrics combined with Shapley values reveal attribute contributions to bias. **FairVis** creates visual clusters of under-representation

05

Theoretical Foundations

Sample complexity bounds establish minimum group sizes required for fair learning guarantees

Identification Methods for Continuous Attributes

Continuous Coverage Analysis

The **Asudeh et al. (2020)** approach extends coverage concepts to numeric data: a point is covered if $\geq k$ neighbors exist within radius ρ.

Geometric visualizations using **Voronoi diagrams** help identify uncovered zones in continuous space (e.g., individuals age > 60 with income < $30K).

Challenges

- Choosing optimal ρ and k involves trade-offs
- Computational complexity for large datasets
- Sensitivity to distance metrics in high dimensions

Step 1: Identify

Detect missing patterns at global and local levels

Step 2: Locate

Determine where bias causes maximum harm

Step 3: Resolve

Apply appropriate mitigation strategies

Critical Understanding: Identification methods answer *where* bias exists and *how severe* it is. This forms the essential first step before any bias resolution strategy or fair model training can begin. Different data types require tailored formulations, but all serve the same goal: building more equitable AI systems.

RESOLVING REPRESENTATION BIAS IN STRUCTURED DATA

we examine 2 cases:

⇒ when we can simply add more data

⇒ when there is no more data available to add

ADDING MORE DATA

- best way to fix underrepresentation is to add more data
- 3 ways - Data Collection, Data Augmentation, Data Integration
- Data Collection - collect as few new samples as possible while covering all MUPs (Minimal Uncovered Patterns)
- Data Augmentation - synthetically generate data (Ex: SMOTE)
- Data Integration combine data from multiple sources

NO MORE DATA TO ADD

- when adding data is impossible, inform users about underrepresentation
- 3 ways - Generating Warnings, Data Labels & Data Sheets, Query
- Generating Warnings - warning triggered when query involves underrepresented regions
- Data Labels & Data Sheets - set of questions dataset creators must answer that will help users assess fitness for purpose before using data
- Query Rewriting - minimally rewrite the transformation queries so that certain representation constraints are guaranteed to be satisfied in the result of the transformation

REPRESENTATION BIAS IN UNSTRUCTURED DATA

we examine 2 cases:

⇒ Image Data

⇒ Natural Language Data

⇒ Graph Data

IDENTIFICATION & RESOLUTION
IN EACH CASE IS STUDIED

IMAGES

IDENTIFICATION

- Reduction to Tabular Data, then apply identification methods for structured data
- Biased Image Representations - represent each image as a low-dimensional vector and measure diversity/clustering
- Cross Dataset Bias Detection - comparing datasets on representativeness
- Crowd-Sourcing - human in the loop by question generation, answer collection and bias judgement

RESOLUTION

- Data Augmentation: add synthetic or transformed samples to increase diversity of dataset (Ex: GANs and sim2real style transfer)
- Reweighting existing samples (Ex: REPAIR with goals of classifier loss and label uncertainty)

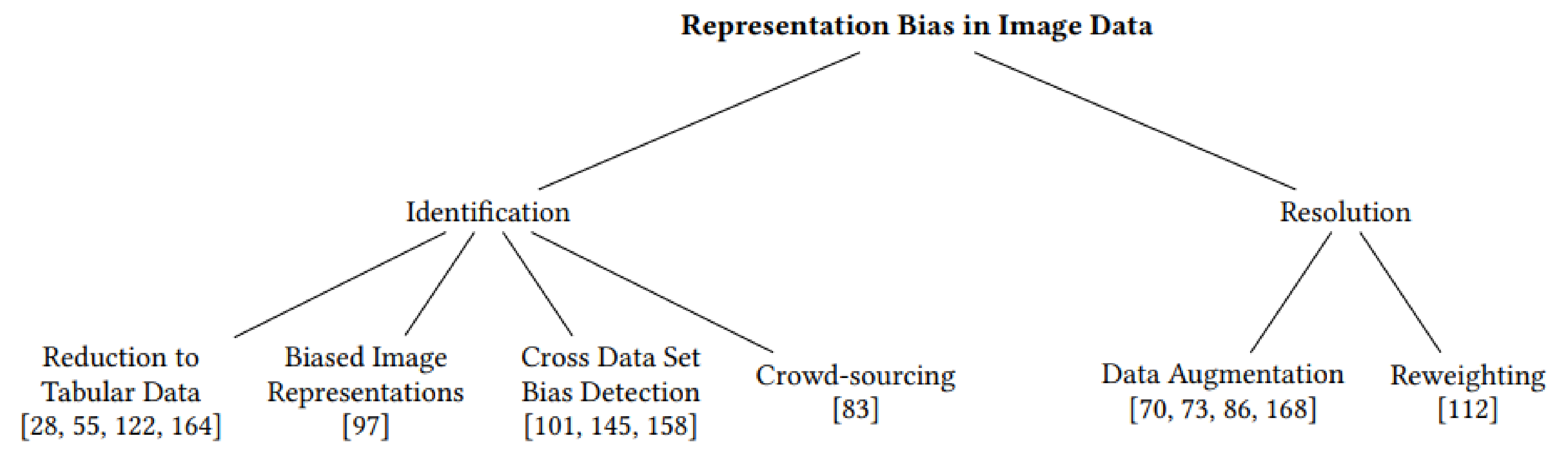

Fig. 8. Classification of techniques on identifying and resolving representation bias in image data sets

TYPES OF REPRESENTATION BIAS IN NATURAL LANGUAGE DATA

- Denigration
- Stereotyping
- Underrepresentation

Each NLP task can involve one or more of these biases.

NATURAL LANGUAGE

IDENTIFICATION

- Performance and Representation Difference Among Sensitive Groups - Gender Swapping Tests & False Positive/Negative Rate Difference between groups
- Analyzing Sub-space Embeddings of Sensitive Attributes: Word embeddings (like Word2Vec or GloVe) analyzed to geometrically detect bias

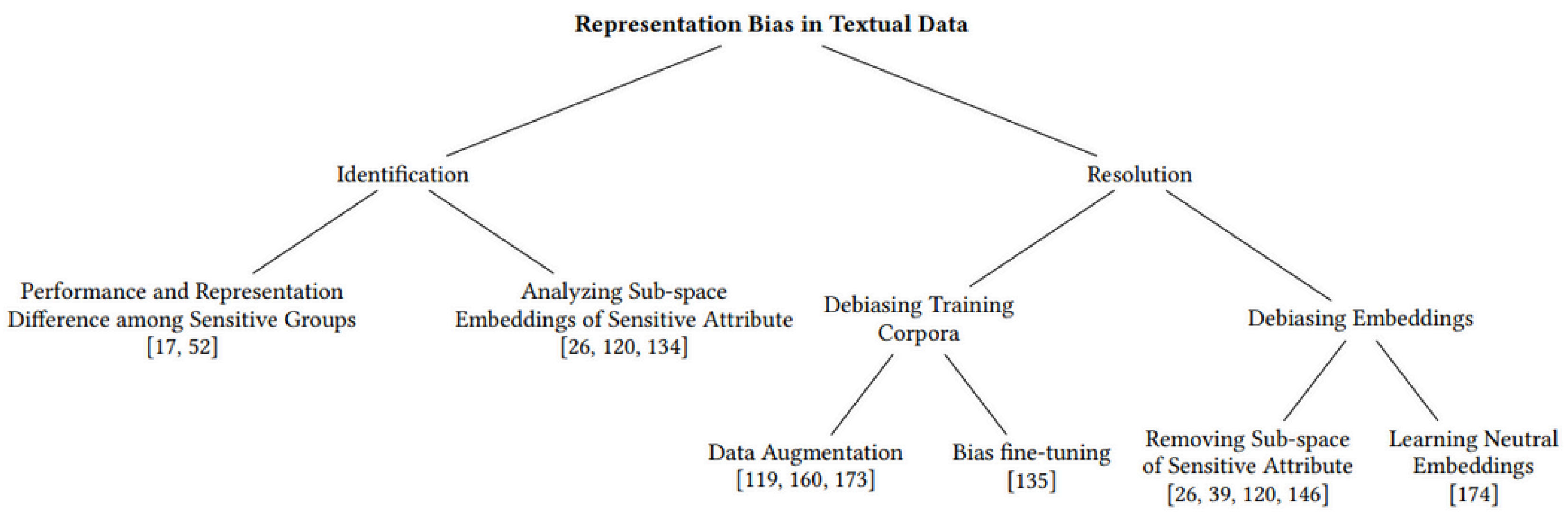

Fig. 9. Classification of techniques on identifying and resolving representation bias in textual data

RESOLUTION

- Text Corpus Alteration - altering the text data itself ⇒ Data Augmentation, Bias Fine-Tuning
- Word Embedding Adjustment - altering the learned representations ⇒ Removing Sub-space of Sensitive Attribute, Learning Neutral Embeddings

In Speech, bias is identified by comparing error rates across subgroups. Demographic info is obtained by manual annotation or automatic classification, and then analysis is similar to tabular data bias identification.

GRAPHS

IDENTIFICATION

- Graph-level Identification - bias in overall structure

$$r = \frac{\sum_i e_{ij} - \sum_i a_i b_i}{1 - \sum_i a_i b_i}$$

where:

$$e_{ij} = \frac{card\{(i,j) \in \mathcal{E}; A_{v_i} = i, A_{v_j} = j\}}{m}, \quad a_i = \sum_j e_{ij} \text{ and } b_j = \sum_i e_{ij}$$

- Embedding-level Identification - bias in learned node embeddings

$$RB = \sum_{a=0}^{l} \frac{1}{|V_a|} \mathrm{AUC}(\{\mathbb{P}_h(a, z_v) | \forall v \in V_a\})$$

RESOLUTION

- Repairing Graph: removing bias by altering the graph's structure (nodes or edges) itself instead of from embeddings (Ex: adjacency matrix)

- Learning Unbiased Embeddings: modify the learning objective to ensure fairness in embeddings (Ex: Fairwalk, CrossWalk & Debayes

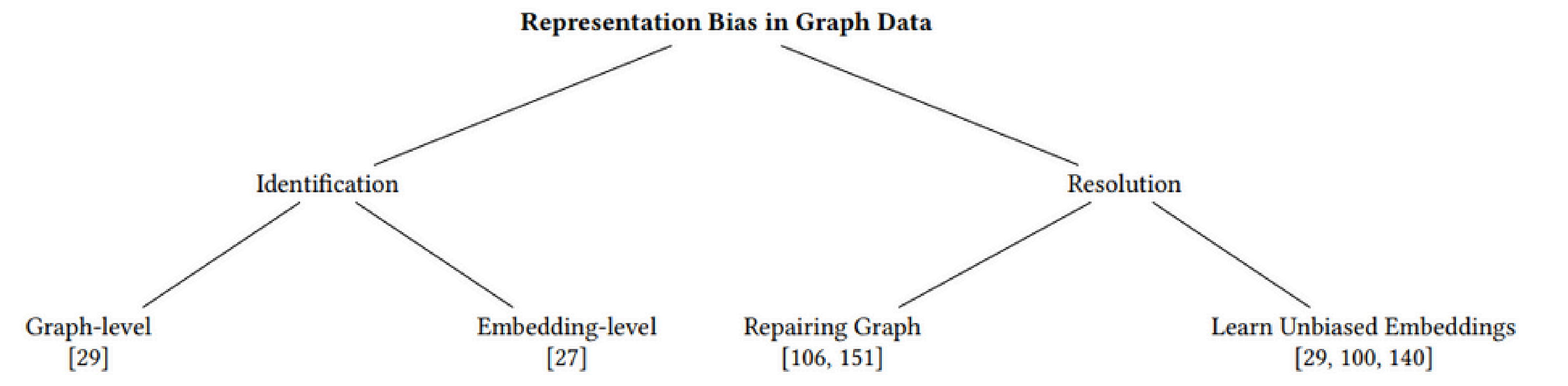

Fig. 10. Classification of techniques on identifying and resolving representation bias in graphs

CONCLUSION

- The survey reviewed a wide range of techniques for identifying and resolving representation bias in both structured and unstructured data.
- The authors offered taxonomies and comparative frameworks for each data type, helping categorize methods based on level (data, model, or embedding) and task (identification vs. resolution).
- The authors emphasize that ensuring data fairness is an ongoing process — there is no final or perfect solution.

THANK YOU!!

ANY QUESTIONS?